

INTRODUCTION

The dominant paradigm in artificial intelligence treats learning as optimization of a predefined objective function over a static dataset. Biological intelligence operates on fundamentally different principles: brains minimize *surprise* through active inference [?], self-organize through Hebbian plasticity and synaptic pruning, and discover causal structure through intervention and counterfactual reasoning.

The gap between these two paradigms is not merely quantitative—it is architectural. Deep learning systems require millions of labeled examples; humans learn causal models from a handful of observations. Reinforcement learning agents maximize engineered reward functions; biological agents minimize *variational free energy*, a quantity that emerges naturally from the imperative to maintain homeostasis in a changing world.

We propose Feynman’s Brain as a bridge between these worlds. It is not a supervised learner, not a reinforcement learning agent in the traditional sense, and not a neuro-symbolic system with handcrafted rules. It is an *evolving causal graph* inhabited by a population of autonomous *cells* whose genome-determined behavior is shaped by a reward signal derived from predictive accuracy—a direct implementation of the free energy principle.

ARCHITECTURE

Knowledge Graph as Generative Model

The foundation of Feynman’s Brain is a directed causal graph $\mathcal{G} = (V, E)$ initialized from a physics knowledge base of 213 physical laws and 603 causal edges spanning 14 domains (classical mechanics, electromagnetism, thermodynamics, quantum mechanics, relativity, etc.). Each edge $u \rightarrow v$ represents a causal relationship: u causes v according to some physical law.

This graph serves as the *generative model* $p(o|z)$ in the free energy framework: given hidden causes z (the causal ancestors of a concept), the graph determines what observations o (the effects) are expected. Unlike probabilistic graphical models, the graph is *discrete* and *deterministic*—edges exist or do not, and traversal follows deterministic paths. Probabilistic behavior emerges not from edge weights but from the population dynamics of cells.

Evolvable Cells

The graph is explored by a population of $N \approx 11,500$ *cells*, each occupying a single node at any time. A cell’s behavior is governed by a real-valued genome $\mathbf{w} \in \mathbb{R}^7$ encoding preferences over seven atomic actions:

Action	Function
step_forward	Walk along an outgoing edge (explore)
step_backward	Walk along an incoming edge (backtrack)
mark	Register a signal at current node
echo	Reinforce existing signals
intervene	Block one edge, walk another (counterfactual)
rest	Consolidate memory
split	Reproduce at a neighboring node

Actions are selected via softmax over genome weights. Crucially, no behavior is hardcoded: a cell with high “mark” weight becomes a registration specialist; a cell with high “intervene” weight becomes an experimenter. Niche differentiation emerges from reward pressure alone.

Free Energy as Reward

The reward signal implements the free energy principle:

$$R(\text{walk}) = \begin{cases} 1.5 \times \text{path_length} & \text{if path is known (prediction hit)} \\ 0.3 \times \text{path_length} & \text{if path is new (prediction miss)} \end{cases} \quad (1)$$

Prediction hits correspond to low free energy (model is correct); prediction misses correspond to high free energy (model must update). This 5:1 ratio drives cells to prefer known causal chains—the system *seeks confirmation* rather than novelty, consistent with the free energy minimization imperative.

Additional reward channels include: registration bonus for first-time path discovery (+0.6), methodology bonus for walking through experiment/verify nodes ($\times 1.4$), and a Eureka multiplier ($\times 4.0$) for long cross-domain known paths.

Synaptic Layer and Sleep

A parallel synaptic layer tracks edge coincidence counts c and connection strengths $s \in [0, \infty)$. Each time a cell traverses an edge, the synaptic weight s is strengthened. During periodic *sleep* cycles (triggered by accumulated walk pressure), all synaptic weights undergo multiplicative decay $s \leftarrow s \times 0.7$, and edges with $s < 0.01$ are naturally eliminated—implementing the synaptic homeostasis hypothesis (SHY) without hard thresholds.

Inheritance Without Normalization

Cell reproduction follows Darwinian principles: a child cell inherits its parent’s genome with small Gaussian mutations ($\sigma \propto \text{mutation_rate}$). Critically, *no normalization*

is applied to the inherited weights. A parent with explore weight 2.0 produces children with explore weight ≈ 2.0 , enabling true genetic propagation of behavioral specialization. This contrasts with earlier versions where weight normalization ($\sum_i w_i = 1$) at birth crushed minority behaviors.

EMERGENT PHENOMENA

Topological Self-Organization

Starting from 213 laws and 603 edges (gen 0), the graph topology evolves dramatically over 1,200 generations (Table ??).

TABLE I. Topological evolution of Feynman’s Brain.

Metric	gen 67,100	gen 69,500	Human Brain
Nodes	7,200	7,523	$\sim 10^{11}$
Edges	28,464	54,395	$\sim 10^{14}$
Max degree	81	174	$\sim 10^{3-4}$
Median degree	5	8	–
Clustering coefficient	0.0071	0.0172	0.1–0.5
Avg. shortest path	4.07	3.51	4–6
Small-world index σ	6.85	8.54	3–10
Hub edge share (top 5%)	24.3%	24.3%	30–50%

The small-world index $\sigma = (C/C_{\text{rand}})/(L/L_{\text{rand}})$ measures how much more clustered the network is compared to a random graph while maintaining short path lengths. $\sigma > 1$ indicates a small-world network; Feynman’s Brain achieves $\sigma = 8.54$, well within the human brain range of $\sigma \in [3, 10]$.

Path Composition: Autonomous Causal Discovery

The most significant emergent capability is *path composition*. When a cell walks $A \rightarrow B$ and $B \rightarrow C$ is a known path, the system composes $A \rightarrow B \rightarrow C$ and registers $A \rightarrow C$ as a new causal edge. This is not a programmed feature—it emerged from the coincidence detection mechanism combined with the compose reward channel.

By gen 69,400, 52,893 composed shortcuts had been created (21% of all emergent edges). Examples include:

- `harmonic_function` $\rightarrow$ `plasma_frequency` (mathematics $\rightarrow$ plasma physics)
- `tidal_force` $\rightarrow$ `cavity_effects` (gravitational $\rightarrow$ quantum)
- `teleportation_fidelity` $\rightarrow$ `length_contraction_factor` (quantum information $\rightarrow$ relativity)

Behavioral Evolution: From Random to Active Inference

At gen 67,100, cell behavior was dominated by mark (61%) with minimal explore (2.8%). By gen 69,500, the distribution had shifted dramatically (Table ??).

TABLE II. Behavioral genome evolution.

Genome weight	gen 67,100	gen 69,500
Mark mean	2.17	1.77
Intervene mean	1.12	1.86
Explore mean	0.0064	0.0125
Explore cells >1.0	6	24
Explore cells >0.5	7	42
Intervene cells >2.0	4,198	7,179

The intervention gene overtook the mark gene for the first time, and the intervention action itself evolved from random edge blocking to expected-information-gain (EIG)-driven selection: cells now *block the least informative edge and walk the most informative alternative*. This is active inference without an explicit active inference module.

Attractor Basin Formation

The coincidence-based synaptic layer naturally forms attractor basins in the knowledge graph. Nodes with high cell populations and strong coincident edges become deep attractor states. The energy landscape $E(n) = -\log(\text{pop}(n)) - \bar{s}(n)$ reveals a structured terrain with distinct basins separated by ridges—the signature of a self-organizing associative memory.

DISCUSSION

Why Not Hierarchical Predictive Coding?

The human brain employs hierarchical predictive coding because its input is a *sensory stream* with natural hierarchical structure (edges $\rightarrow$ textures $\rightarrow$ objects $\rightarrow$ scenes). Feynman’s Brain operates on a *causal graph*—there is no sensory hierarchy, only causal depth. Imposing hierarchical modules would constitute *method injection* rather than *capacity provision*.

Instead, the system uses topological depth (BFS from root concepts) as a natural proxy for abstraction level. Cells differentiate into deep-node specialists (theory-focused) and shallow-node specialists (observation-focused) through niche competition alone.

Deterministic vs. Probabilistic Computation

Feynman’s Brain is fundamentally deterministic: edges exist or do not, c-values are integer counts, and rewards are fixed ratios. Yet probabilistic behavior emerges at the population level through the softmax action selection and the synaptic strength continuum $s \in [0, \infty)$. This mirrors the biological brain where individual synapses are stochastic but population codes are robust.

The transition from c-value-based hard pruning ($c < 2 \Rightarrow \text{delete}$) to s-value-based soft decay ($s \times 0.7$, natural elimination at $s < 0.01$) represents a shift from deterministic to probabilistic edge maintenance—one that significantly improved attractor stability.

Autonomous Scientific Discovery

To our knowledge, Feynman’s Brain is the first system to demonstrate autonomous causal discovery through self-organized evolution on a physics knowledge graph. The composed shortcuts represent genuine hypotheses about causal relationships that the system has *inferred* rather than been *told*. Whether these hypotheses correspond to real physical laws requires external verification—the system can propose, but not yet validate, scientific claims.

The next frontier is closing the loop: using the intervention mechanism to *test* composed hypotheses and updating confidence based on experimental outcomes. This would implement the complete scientific method within a single autonomous agent.

CONCLUSION

Feynman’s Brain demonstrates that autonomous causal discovery can emerge from three simple principles: (1) a knowledge graph as generative model, (2) a population of evolvable cells minimizing prediction error, and (3) coincidence-driven synaptic plasticity with sleep-mediated homeostasis. The system requires no labeled data, no reward engineering beyond the free energy signal, and no hardcoded behavioral strategies. Yet it spontaneously develops small-world topology ($\sigma = 8.54$), composes 53,000 causal shortcuts, evolves active inference behavior, and forms structured attractor basins—all emergent properties of a brain that was given capacity but not method.

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